A Context-Aware Self-Healing Security Framework for WBAN Against Sybil Attacks

Final Year Project Group - 15

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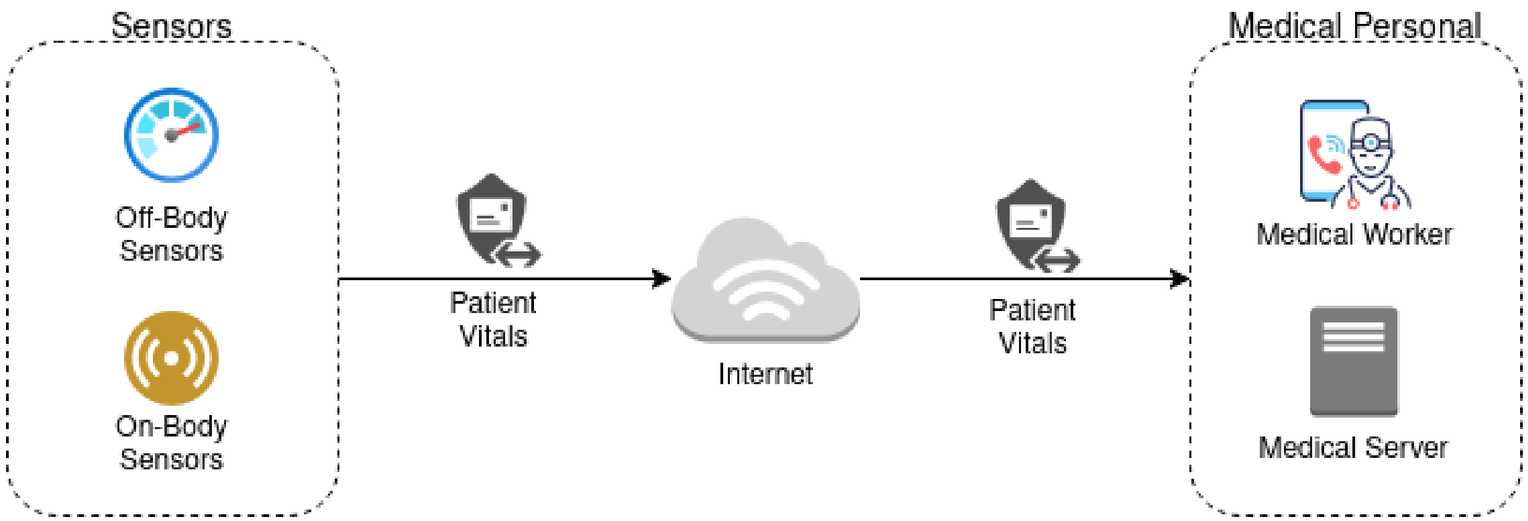
1. INTRODUCTION

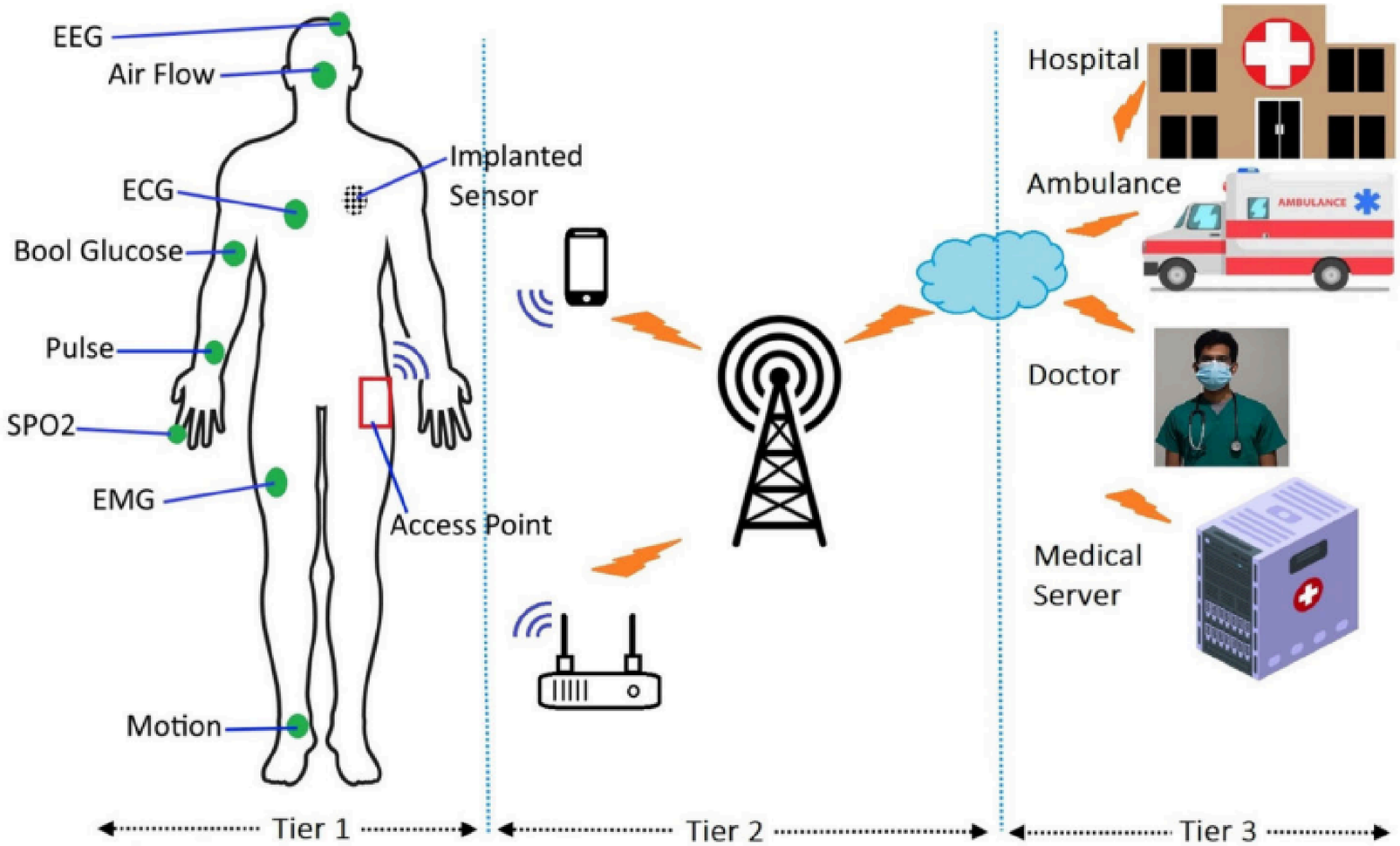
What is IoMT?

- Network of medical devices, sensors, software, and healthcare systems
- Connected through the internet.
- Used to improve patient monitoring, diagnostics, and treatment.



1. INTRODUCTION CONT.

What is WBAN?

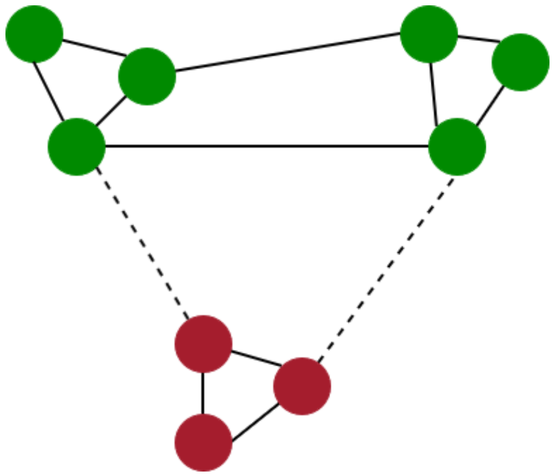


2. SECURITY IN WBAN

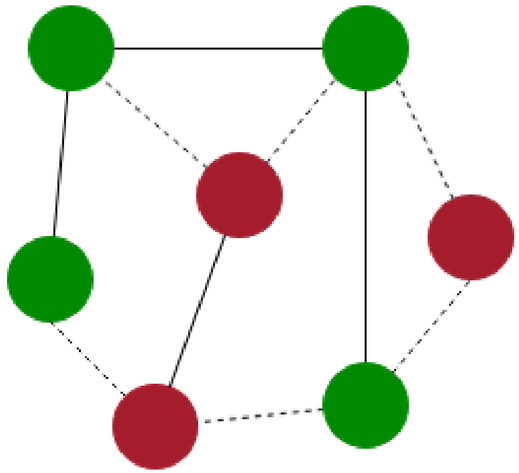
- Life-critical and highly sensitive data.
- Tiny, low-power devices resulting weak security.
- Open wireless channels making it easy to eavesdrop & spoof.
- Gateway to larger IoMT networks.
- Attacks:
    - Sybil attacks
    - Sinkhole attacks
    - Selective forwarding
    - Replay attacks
    - DoS attacks

3. WHAT IS A SYBIL ATTACK?

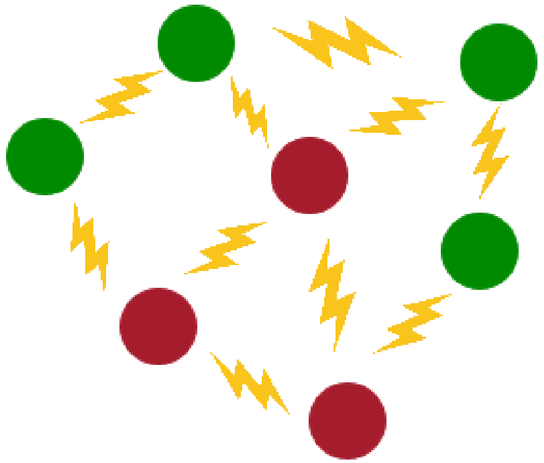
- A single malicious node creates multiple fake identities.



SA-I



SA-II



SA-III

4. WHY SYBIL?

One fake node can mimic dozens of sensors.

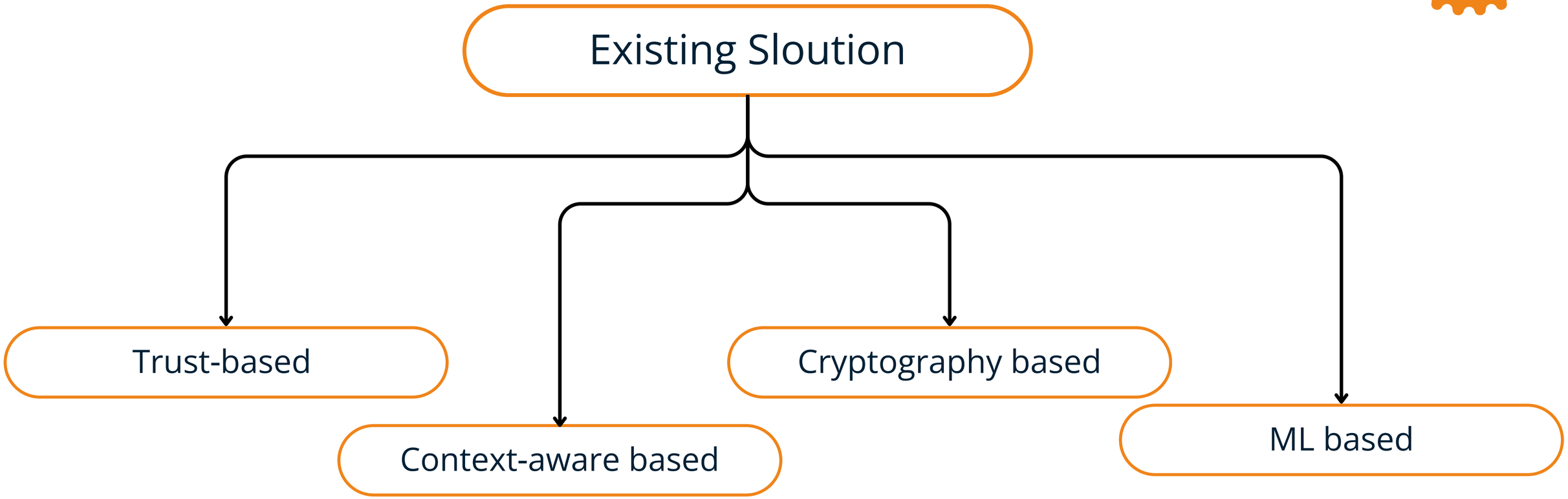
Can manufacture false data, enabling secondary attacks.

Traditional defenses are heavy and not optimized for WBAN.

5. GOALS AND OBJECTIVES

Focused on Sybil attack detection and mitigation approaches in IoMT based WBA Network

6. EXISTING SOLUTIONS



6. EXISTING SOLUTIONS

Trust-based

6. EXISTING SOLUTIONS CONT.

Approach	Contribution	Limitation
BFT-IoMT	Decentralized blockchain-based trust framework verification and network behavior, energy	Limited scope Bad-Mouthing Attack Ballot-Stuffing Attack
FTM-IoMT	Centralized Based on Integrity and Compatibility Detecting framework	Less Scalability static fuzzy rule
LETM-IoT	All type of sybil based on parent child relationship, sequence ID Trust based	Centralized Dependency Not specific for IOMT
TRUST BASED TWO PHASE ALGO	based on suspicious comm. behavior neighbor node trust values Isolate sybil node	threshold dependency Bad-Mouthing Attack

6. EXISTING SOLUTIONS CONT.

- Techniques: Network Behavior-Based Detection, Threshold-Based Decision Making
- Strength: Lightweight Algorithm Design, Energy and Resource Efficiency
- Limitation: Simulation-Only Validation, Static Thresholds and Rules, lack of adaptivity, Bad-Mouthing Attack and Ballot-Stuffing Attack

6. EXISTING SOLUTIONS

Context-aware based

6. EXISTING SOLUTIONS CONT.

Approach	Contribution	Limitation
SAPA-BAN	Uses behavioral context (ID patterns, path formation behavior) to detect attack and re initiates path	Vague Detection Mechanism built on asusmption
RSSI-Context Shield	Received signal strength indicator physical for detection	Unreliable detection
Adaptive Context Analyzer (Kuraku)	Continuously updates its ML model using contextual threat patterns and repository data;	Vague methodology with undefined internal components; unrealistic performance claims;
Mobile IoT sybil Detection	RSSI behavior, two round detection with RSSI range using intelligent algo (LSTM/GAN)	Limited Testing Scope Dependence on Accurate Position Prediction

6. EXISTING SOLUTIONS CONT.

- Techniques: Identity Context, Behavioral Context, Network Context, Environmental Context, User Physiological Context
- Strength: Reduced False Positives, Better Adaptability to Dynamic Networks
- Limitation: Not specific to sybil attack, Dependency on Assumptions

6. EXISTING SOLUTIONS CONT.

Cryptographic-based

6. EXISTING SOLUTIONS CONT.

Approach	Contribution	Limitation
Rechained-CryptoID Framework	Block-chained based identity system Digital identity and revocation	High computational + energy requirements for small IoT devices.
SybilGuard	Provides lightweight 64-bit block cipher ID, key, auth token authentication for IoT nodes	encryption weakness single point of failure
RSSI-Context Shield	Based on RSSI for detection CCA , lightweight encryption, AODV	weak encrpt, unreliable detection
Crypto-Based Identity & Encryption Framework	Ensures secure device identity using cryptographic keys; Provides integrity + authentication for IoT nodes	High cryptographic cost for small IoT devices; Increased energy + computational requirements

6. EXISTING SOLUTIONS CONT.

- Techniques: Lightweight Encryption, Access/Identity Validation
- Strength: Lower Resource Usage, Defense in Layers
- Limitation: Limited Real-World Reliability, Lack of Strong Validation, Bypassable Under Smart Attackers, Cryptographic Weakness

6. EXISTING SOLUTIONS CONT.

Machine Learning based

6. EXISTING SOLUTIONS CONT.

Approach	Contribution	Limitation
Sybil in WSN with Supervised Lea.	IDS Network traffic-based Four ML based attack detection	Not real time lack of content awareness dataset limitation
Advancements in Sybil Detection	based on network traffic, signal, node behavior detecting with supervised, semi-supervised	lack of content awareness Adaptation Needs Not specific in IOMT
Health Guard	based on Vital sign, user activity context medical behaviors ML algo	Real world deployment Not specific for sybil
Deep Learning-based IDS for IOMT	Network flow features an biometric sensor data CNN+LSTM	Not specific for sybil High computation power

6. EXISTING SOLUTIONS CONT.

- Techniques: Traffic-based , Node Behavioral-based, semi-Supervised ML, Fog-Cloud Integration
- Strength: Adaptive & Flexible
- Limitation: Simulation-Only Validation, Lack of Real-time Self-Healing, Limited Context Awareness

7. RESEARCH GAP

- Absence of Context Awareness
- No Autonomous Recovery / Self-Healing
- Low Scalability and Practical Deployability
- Lack of Lightweight and Real-Time Sybil Defense for WBAN
- No standardized WBAN dataset for Sybil patterns

8. PROBLEM STATEMENT

- IoMT-based WBANs lack a lightweight,
  - Context-aware framework
  - Autonomously recovering from Sybil attacks

- This leads to compromised patient safety and unreliable medical decisions.

9. RESEARCH QUESTION

How to detect Sybil attacks in WBAN systems and recover without increasing energy consumption or latency?

10. PROPOSED SOLUTION

- A hybrid lightweight security framework
- Combines cryptographic identity verification with context-aware machine learning to,
  - Detect Sybil attacks
  - Minimize false positives
  - Automatically self-heal compromised WBAN nodes

11. METHODOLOGY

Phase 01 - Literature Review

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Phase 02 - Dataset Preparation

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Phase 03 - ML Model Development

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Phase 04 - Self-Healing Mechanism

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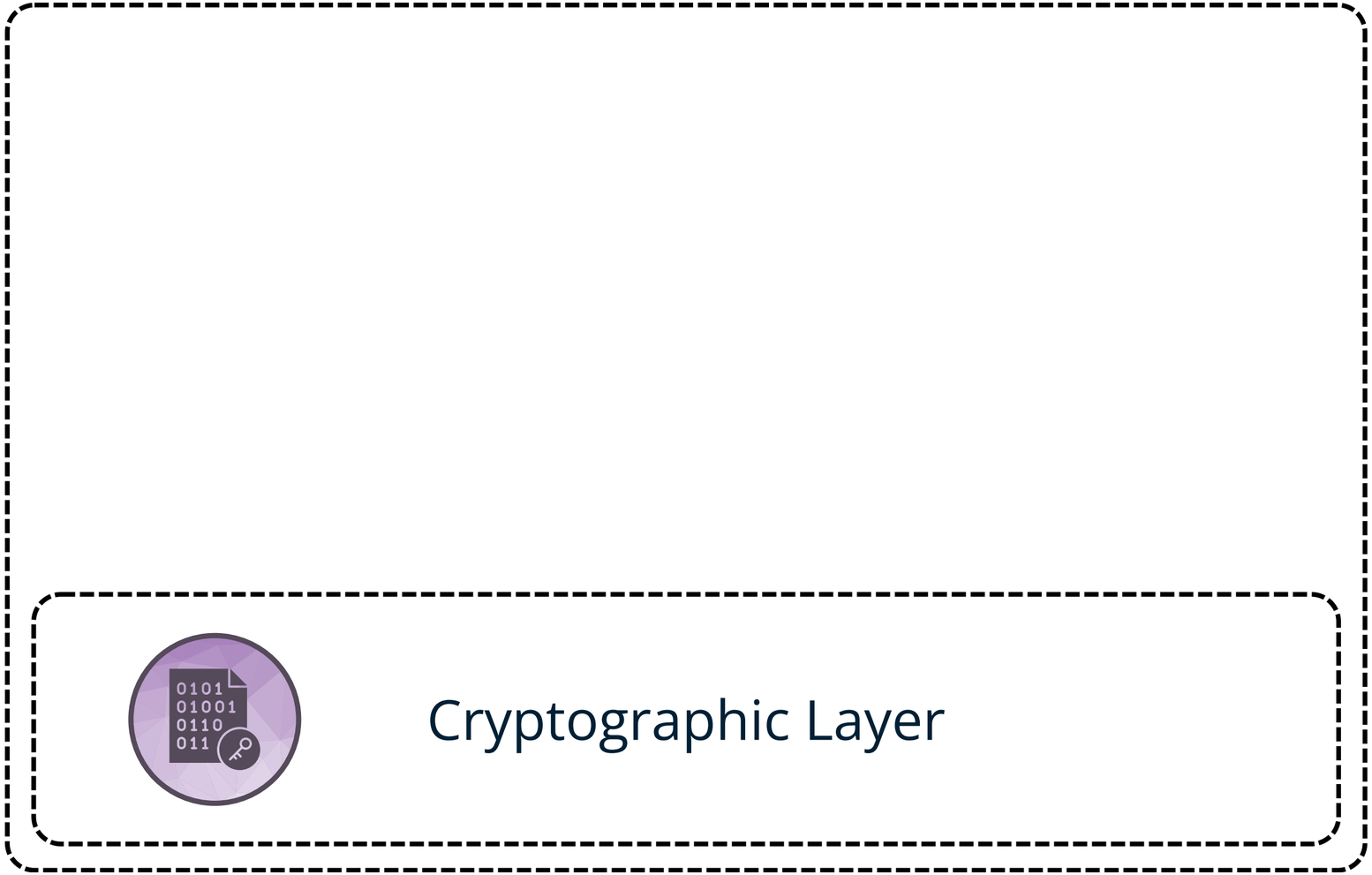
Phase 05 - Evaluation

12. FRAMEWORK ARCHITECTURE



12. FRAMEWORK ARCHITECTURE CONT.

Cryptographic Layer

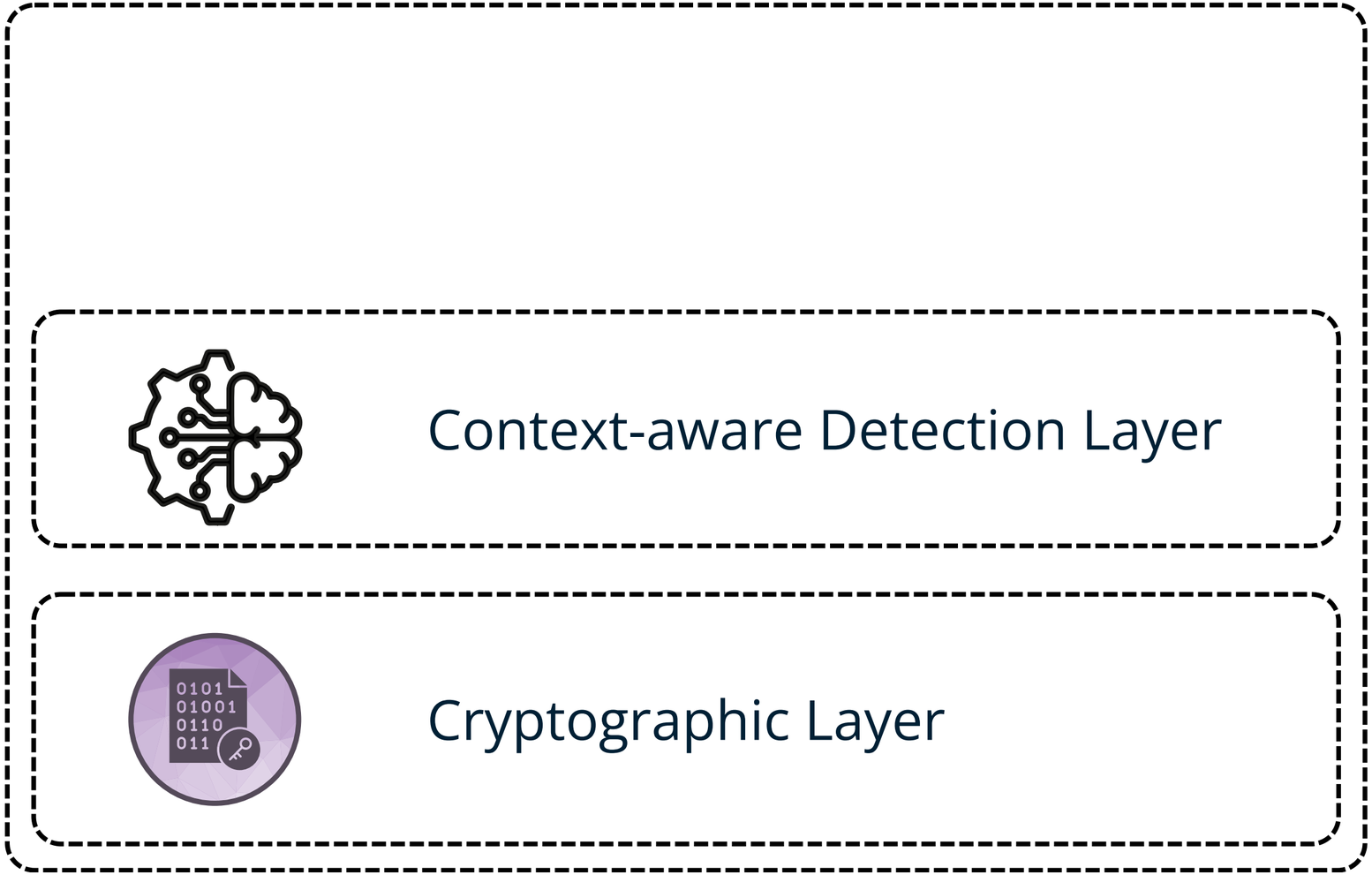
Methods

- Pre-Shared Symmetric Keys
- PUF-Derived Key
- Secure Element Stored Keys
- ECC (Public/Private Keys)

SensorID	Data	Timestamp

SensorID	Data	Timestamp	Signature	PublicKey

12. FRAMEWORK ARCHITECTURE CONT.

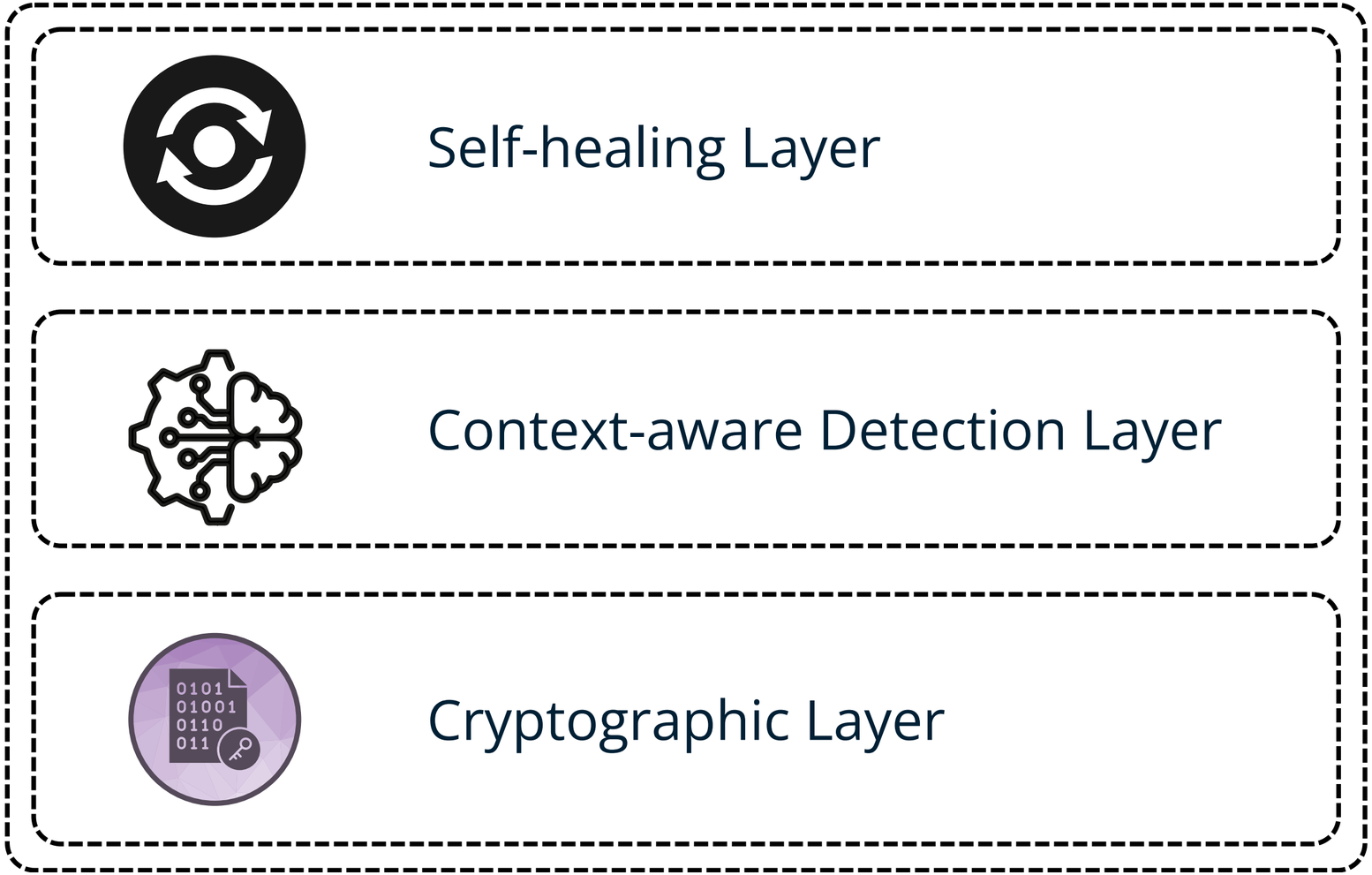


12. FRAMEWORK ARCHITECTURE CONT.

Context-aware Detection Layer

Network Context Aware	Medical Context Aware
• RSSI Variation (Received Signal Strength Indicator) • Packet Transmission Rate • Link Quality Indicator (LQI)	• Posture and Movement Awareness • Physiological Signal Plausibility • Cross-Sensor Correlation • Temporal Behavior of Vitals • Recovery Patterns

12. FRAMEWORK ARCHITECTURE CONT.

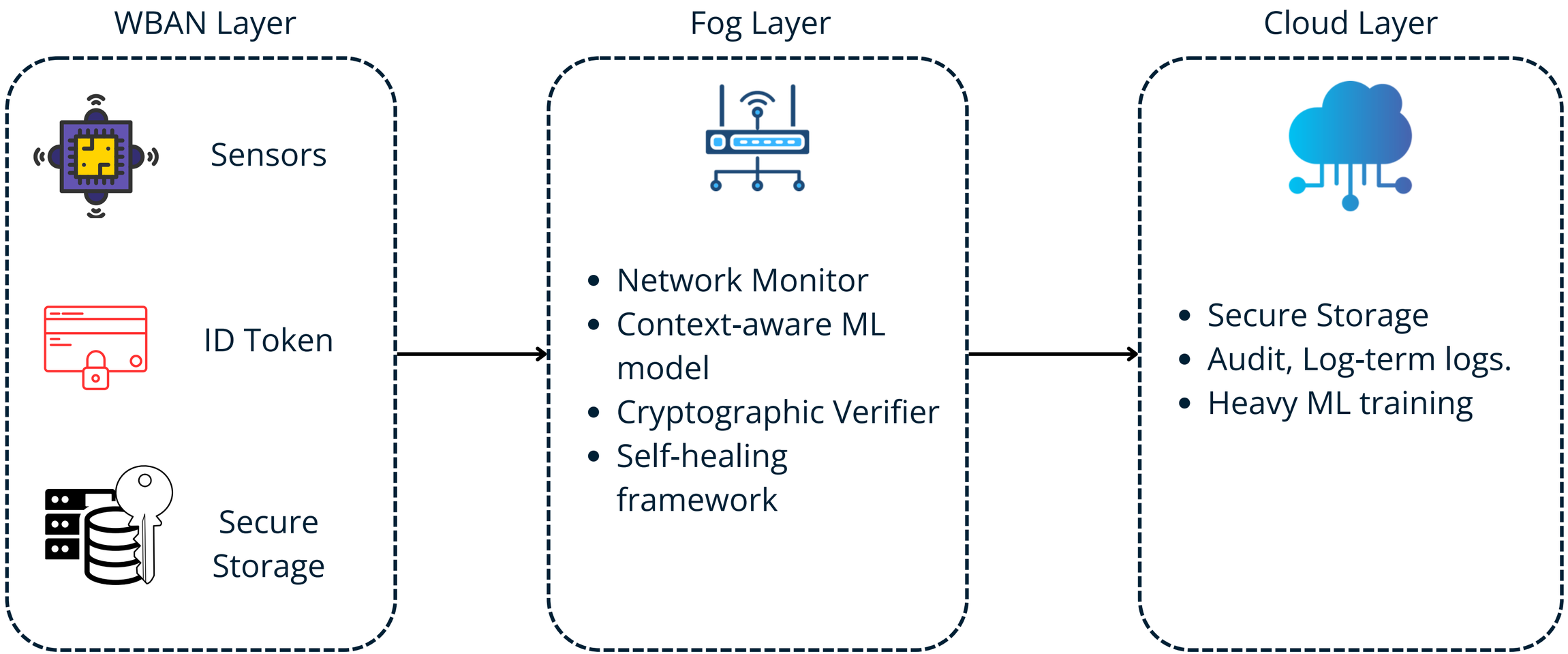


12. FRAMEWORK ARCHITECTURE CONT.

Self-healing Layer

- Automatic isolation of suspicious nodes
- Revoke security keys
- Adaptive threshold adjustment
- Reclassification
- Fail-Safe mode activation

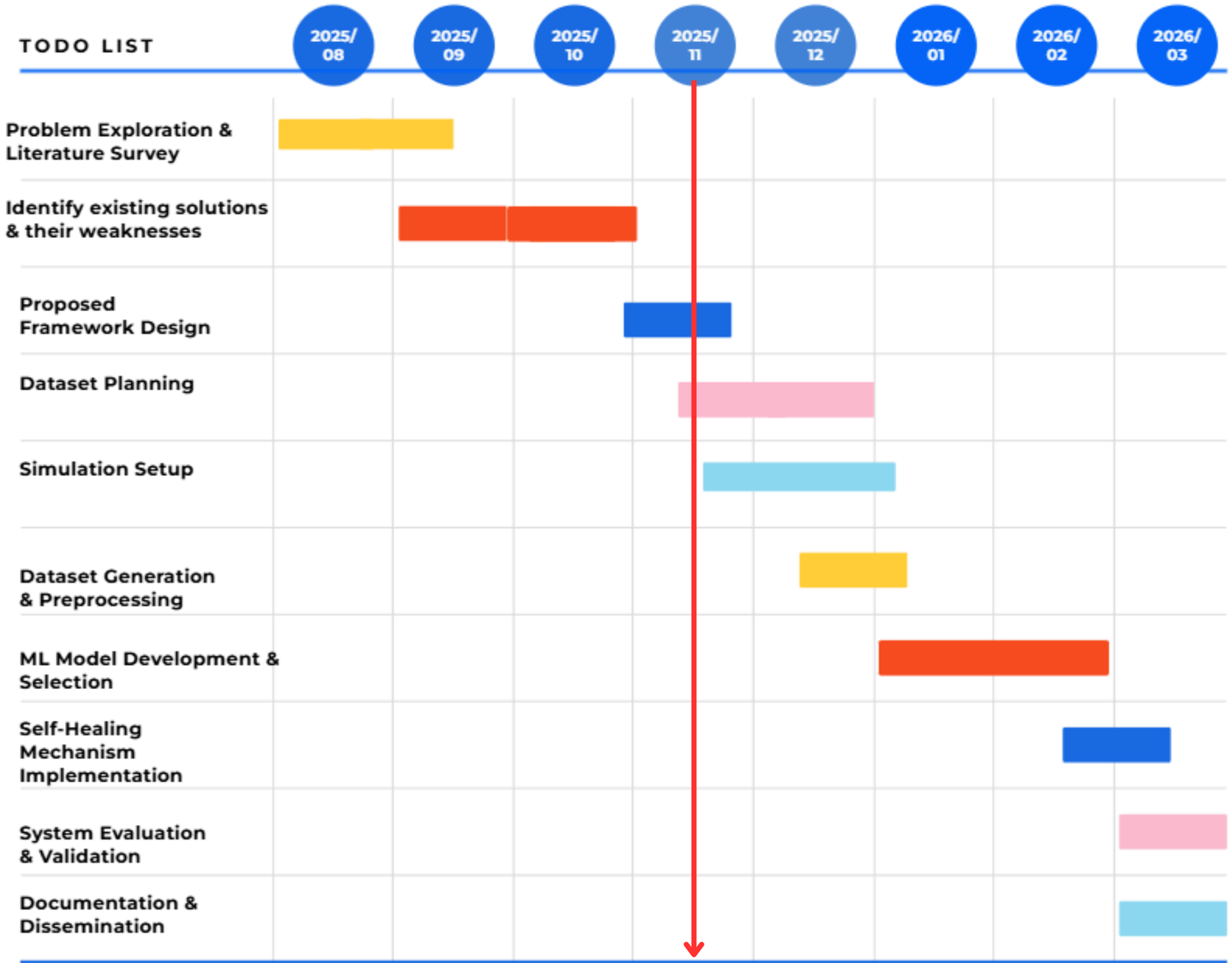
13. HIGH-LEVEL ARCHITECTURE



14. EXPECTED OUTCOMES/ DELIVERABLES

- A simulated dataset for Sybil attacks
- A context-aware feature extraction module
- A lightweight Sybil detection ML model
- A self-healing mechanism for WBAN

15. TIMELINE


TODO LIST
2025/ 08
2025/ 09
2025/ 10
2025/ 11
2025/ 12
2026/ 01
2026/ 02
2026/ 03
Problem Exploration & Literature Survey
Identify existing solutions & their weaknesses
Proposed Framework Design
Dataset Planning
Simulation Setup
Dataset Generation & Preprocessing
ML Model Development & Selection
Self-Healing Mechanism Implementation
System Evaluation & Validation
Documentation & Dissemination

Thank You !

Q & A

REFERENCE